Finite Groups and Their Representations — Exercise Sheet 6

Characters and their constructions

Variant: Questions only

Exercise 1 (Hom as a tensor product). For finite-dimensional complex representations U, V of a finite group G, construct an explicit isomorphism $U^* \otimes V \cong \mathrm{Hom}(U, V)$, with the actions of Lecture 6. For the irreducible plane E of S_3, use characters to decompose $E \otimes E$ and $\mathrm{End}(E)$. In the second space G acts by conjugating operators.

Exercise 2 (Squares of the standard representation of S_4). Let $W = \{x \in \mathbb{C}^4 : \sum_i x_i = 0\}$ and let F be the lift of E from $S_4/V_4 \cong S_3$. Use Lecture 6's character table to compute and decompose $\mathrm{Sym}^2(W)$, $\Lambda^2(W)$ and $W \otimes W$. Give each character on types (1^4), $(2, 1^2)$, (2^2), $(3, 1)$, (4).

Exercise 3 (Equal tables for different groups). Build the character tables of D_4 and Q_8 and match their columns. For D_4 use representatives e, r^2, r, s, rs. For Q_8 use $\mathbb{1}, -\mathbb{1}, a, b, ab$, where a, b are the matrices on Sheet 2. For $\varepsilon, \eta \in \{1, -1\}$, write $\chi_{\varepsilon,\eta}$ for the linear character taking r, s to ε, η in D_4, or a, b to ε, η in Q_8. A + or − in a character subscript means 1 or -1. Show that both have four linear characters and one plane, with identical tables under this correspondence. Explain why the groups are not isomorphic. In each group call the plane module P, with character χ_P.

Exercise 4 (The general dihedral tables and their kernels). For $n \geq 3$, use the representations from Sheet 5 and the classes from Sheet 1 to give the character table of D_n, separately for odd and even n. Include class sizes and parameter ranges. For an irreducible plane ρ_k, find its kernel and all g with $|\chi_k(g)| = 2$. State when it is faithful and illustrate with $n = 12, k = 2$.

Exercise 5 (Conjugacy to the inverse). Prove that if every element of a finite group G is conjugate to its inverse, then every complex character of G is real-valued. Apply this to S_n. Contrast with a nontrivial linear character of C_3.

Exercise 6 (Normal subgroups as intersections of character kernels). Let G be a finite group and $N \trianglelefteq G$. Prove that

$$N = \bigcap_{\substack{U \in \mathrm{Irr}(G) \\ N \leq \ker \chi_U}} \ker \chi_U .$$

For $G = D_6$ and $N = \langle r^3 \rangle$, list these irreducibles using Exercise 4. Find the smallest dimension of a complex representation of G whose kernel is exactly N.

Exercise 7 ⋆. For the plane P in each group of Exercise 3, compare the characters of $\mathrm{Sym}^2(P)$ and $\Lambda^2(P)$ under the common character-table column identification. Does an abstract character table determine these constructions? Use $\chi_{\varepsilon,\eta}$ for the linear rows defined there.